

# SCUPI 2025S: Microelectronics Circuits

## Lab #2 Assignment: Diode Limiting Circuits

### Objectives

- Measure and design limiting & protection circuits.

### Useful links

Diode model number: 1N4007-TP

Diode datasheet: [https://www.mccsemi.com/pdf/Products/1N4001~1N4007\(DO-41\).pdf](https://www.mccsemi.com/pdf/Products/1N4001~1N4007(DO-41).pdf)

### Introduction

In the lecture, we learned limiting & protection circuits and inverting amplifiers. In this lab, we will measure and design limiting & protection circuits. Please perform the following actions and submit your report using the format at the end of this documentation.

**Action #1.1:** Analyze Figure 1 and draw analyzed transfer characteristics and output voltage in the time domain, using the constant-voltage-drop model. Assemble the circuit shown in Figure 1. Provide 14 V<sub>PP</sub> at 1 kHz for V<sub>IN</sub>. Use the oscilloscope to obtain a simultaneous display of V<sub>IN</sub>(t) and V<sub>OUT</sub>(t). Take a picture of your circuit and equipment. Include a figure to explain your setup using the picture. Annotate it with arrows and texts in the figure. Include a captured image of the oscilloscope and discuss the result. Use the oscilloscope in the X-Y mode to display the transfer characteristics of the circuit in a different way. Include a captured image of the oscilloscope and discuss the result.

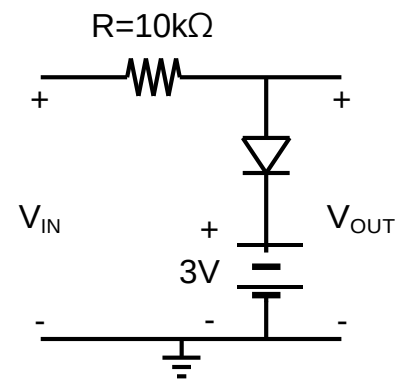


Figure 1

**Action #1.2:** Design a circuit to achieve the transfer characteristics shown in Figure. Draw the schematic and explain its operation, using the constant-voltage-drop model. Assemble the circuit and provide 14 V<sub>PP</sub> at 1 kHz for V<sub>IN</sub>. Use the oscilloscope to obtain a simultaneous display of V<sub>IN</sub>(t) and V<sub>OUT</sub>(t). Take a picture of your circuit and equipment. Include a figure to explain your setup using the picture. Annotate it with arrows and texts in the figure. Include a captured image of the oscilloscope and discuss the result. Use the oscilloscope in the X-Y mode to display the transfer characteristics of the circuit in a different way. Include a captured image of the oscilloscope and discuss the result.

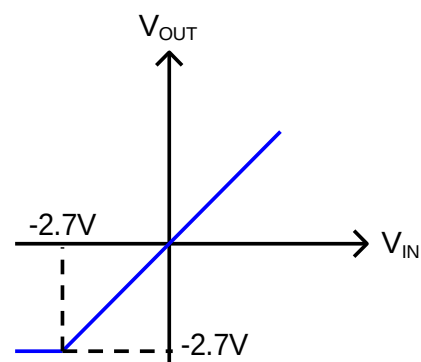


Figure 2

**Action #1.3:** Design a circuit to achieve the transfer characteristics shown in Figure 3. Draw the schematic and explain its operation, using the constant-voltage-drop model. Assemble the circuit and provide 14 V<sub>pp</sub> at 1 kHz for  $V_{IN}$ . Use the oscilloscope to obtain a simultaneous display of  $V_{IN}(t)$  and  $V_{OUT}(t)$ . Take a picture of your circuit and equipment. Include a figure to explain your setup using the picture. Annotate it with arrows and texts in the figure. Include a captured image of the oscilloscope and discuss the result. Use the oscilloscope in the X-Y mode to display the transfer characteristics of the circuit in a different way. Include a captured image of the oscilloscope and discuss the result.

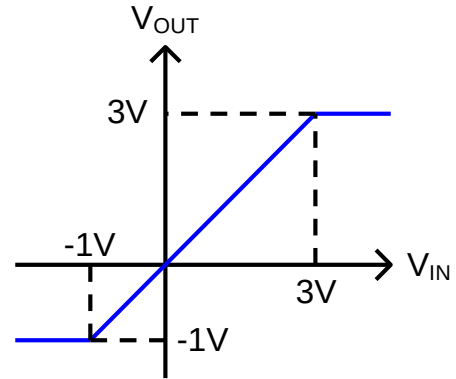


Figure 3

## SCUPI 2024S: Microelectronics Circuits

### Lab #2 Report: Diode Limiting Circuits

#### Action #1.1

Analyze Figure 1 and draw analyzed transfer characteristics and output voltage in the time domain, using the constant-voltage-drop model.

Take a picture of your circuit and equipment. Include a figure to explain your setup using the picture.

Include a captured image of the oscilloscope that shows  $V_{IN}(t)$  and  $V_{OUT}(t)$  and discuss the result.

Include a captured image of the oscilloscope that displays the transfer characteristics of the circuit and discuss the result.

**Action #1.2**

Draw the schematic and explain its operation, using the constant-voltage-drop model.

Take a picture of your circuit and equipment. Include a figure to explain your setup using the picture.

Include a captured image of the oscilloscope that shows  $V_{IN}(t)$  and  $V_{OUT}(t)$  and discuss the result.

Include a captured image of the oscilloscope that displays the transfer characteristics of the circuit and discuss the result.

**Action #1.3**

Draw the schematic and explain its operation, using the constant-voltage-drop model.

Take a picture of your circuit and equipment. Include a figure to explain your setup using the picture.

Include a captured image of the oscilloscope that shows  $V_{IN}(t)$  and  $V_{OUT}(t)$  and discuss the result.

Include a captured image of the oscilloscope that displays the transfer characteristics of the circuit and discuss the result.